

Malik AlNajjar

Software Engineer · Full-Stack & Data

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SUMMARY

Software Engineer at MicroMobility in Jeddah, running the production booking, check-in and point-of-sale platform behind its group rides (PostgreSQL/Supabase, JavaScript, Cloudflare). B.Sc. Software Engineering, UBT, First Class Honours (GPA 4.97/5.0); AWS-certified in data engineering and solutions architecture, with projects across IoT, machine learning and data quality. Bilingual Arabic/English (IELTS 7.5).

EXPERIENCE

- Software Engineer** — MicroMobility, Jeddah Jun 2026 – Present
- Build and run the company's booking and operations platform: customer booking with Google/Apple sign-in, staff check-in, point of sale, inventory and refunds, used every ride night.
 - Model the backend in PostgreSQL on Supabase: row-level security on every table, staff-only functions, realtime sync across staff devices, 85+ versioned migrations.
 - Ship NFC bike check-in, Apple Wallet passes and a 10-language interface (Arabic and Urdu right-to-left).
 - Guard releases with a Playwright end-to-end suite (database stubbed) and GitHub Actions checks for lint, types and translations.
- Rights Protection Officer** — AFC Local Organising Committee, KSA Apr 2026 – Present
- Safeguard AFC intellectual property and broadcast rights at the AFC Champions League Elite Finals 2026; previously Brand Protection Member, AFC Asian Cup U-23 (Jan 2026).

PROJECTS

- AUJ** — Smart Retail Shelf System · Senior Capstone, UBT 2025 – 2026
- ESP32-S3 · MQTT · FastAPI · PostgreSQL 16 · LightGBM · YOLOv12n · WebSocket · React**
- Dual-sensor (load cell + camera) IoT pipeline: 10 Hz weight state machine on ESP32-S3, MQTT ingestion into PostgreSQL 16, 29 REST endpoints and a live WebSocket layer on FastAPI; pick-to-dashboard latency under 5 seconds against a 60-second requirement.
 - Per-SKU demand forecasts with LightGBM quantile models (q10/q50/q90, M5 data), a 5-rule audited pricing engine (±15% band) and YOLOv12n cross-checks of every event; zero PII, 143-test pytest suite on real PostgreSQL.
- Sadeed** — Data Quality & Validation Platform · GASTAT Hackathon, National Finalist 2026
- FastAPI · Redis · Great Expectations · One-Class SVM · Qwen 2.5 (Ollama) · Browser Extension**
- Real-time 3-layer validation of Arabic government survey data (Great Expectations, One-Class SVM, and a local LLM that explains each conflict): detection up from 23% to 87.5% (F1 0.829 vs 0.374), over 97% precision, no data leaves the government network.
 - Code: github.com/malikaalnajjar/Sadeed_GASat_Hackathon
- Tayseer** — Autonomous Smart Wheelchair · AVITHON 2025 (GACA), Top 15 of 500+ teams 2025
- Jetson Nano · ROS · LiDAR · Intel RealSense · TLS 1.3 · AES-256**
- Data scientist on a five-member team: LiDAR, depth-camera and wheel-encoder fusion on Jetson Nano edge AI, paired with a travel-assistant app; security by design; need validated through field research at King Abdulaziz International Airport.
- Masra** — Smart Adventure Tourism Platform · Aseer Tabtaker 2025 2025
- AI developer (team of five): Flutter + Mapbox booking app, AWS IoT Core early warning, scikit-learn models on SageMaker.

SKILLS

Web & Backend: JavaScript, TypeScript, PostgreSQL (PL/pgSQL, RLS), Supabase (Auth, Realtime), Cloudflare Pages & Workers, PWA / Service Workers, IndexedDB, REST & WebSocket, FastAPI, React

Data & ML: Python (Pandas, NumPy, scikit-learn), SQL, Great Expectations, Redis, LightGBM, One-Class SVM, YOLO / ByteTrack, Ollama / local LLMs, Power BI (DAX, Power Query)

Cloud, DevOps & Testing: AWS, Docker, GitHub Actions, Playwright, pytest, Git, Linux / Bash · **IoT & Edge:** ESP32-S3, MQTT, Jetson Nano, ROS, LiDAR, RealSense

Also trained in: Apache Kafka, dbt, Apache Iceberg, AWS Glue / Athena / Lambda, Tableau, QuickSight, R

EDUCATION

B.Sc. Software Engineering — University of Business and Technology (UBT), Jeddah · GPA 4.97/5.0, First Class Honours 2026

KAUST Advanced AI Specialization: machine learning, deep learning, Python Jan – Feb 2025

CERTIFICATIONS

AWS Certified Data Engineer – Associate (DEA-C01, Mar 2026) · AWS Certified Solutions Architect – Associate (SAA-C03, Feb 2026) · AWS Certified Cloud Practitioner (CLF-C02, Dec 2025) · DAMA CDMP (Jan 2026) · PMI CAPM (Jan 2026) · CompTIA Data+ (Nov 2025) · IBM Data Engineering Professional Certificate (Dec 2025) · Google Data Analytics (Dec 2025) · Google Project Management (Jan 2026) · DataCamp Associate Data Engineer in SQL (Jun 2025) · McKinsey Forward Program (Jul 2025)

AWARDS & LEADERSHIP

- Top 15 of 500+ teams, AVITHON 2025 (GACA) · National Finalist, GASTAT Hackathon (2026) · 2nd Place, Project Management Championship by Beyond (Dec 2025)
- Outreach Team Leader, GDG On Campus UBT (Oct 2025 – Jun 2026; 100+ students, 40%+ attendance growth) · Vice Chair & Treasurer, ACM UBT (Feb – Dec 2025; SAR 5,000 budget, 3–5 events per cycle)
- Volunteer: Falling Walls Foundation (Sep 2025) · APCG 2026 international conference at UBT (Feb 2026)